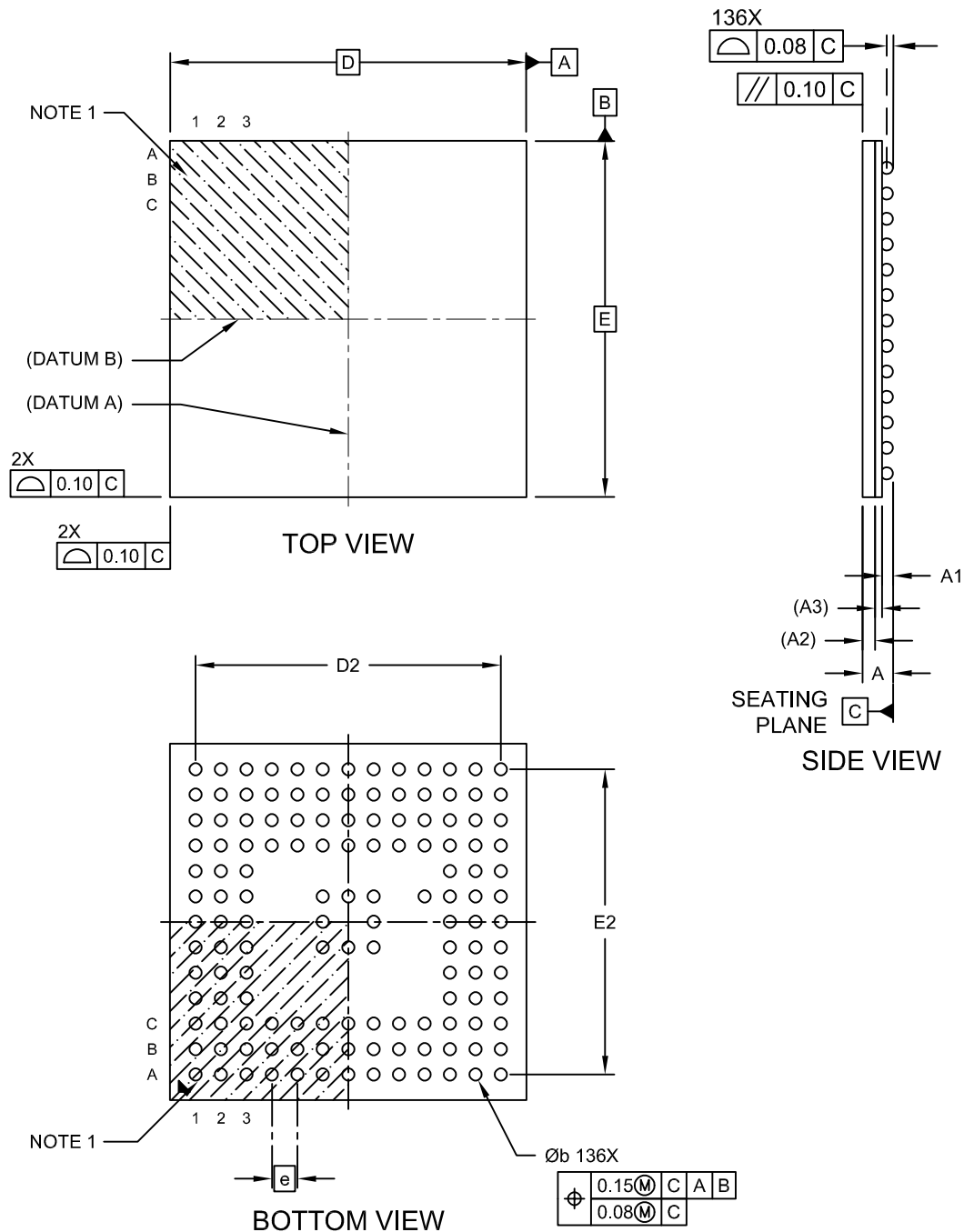


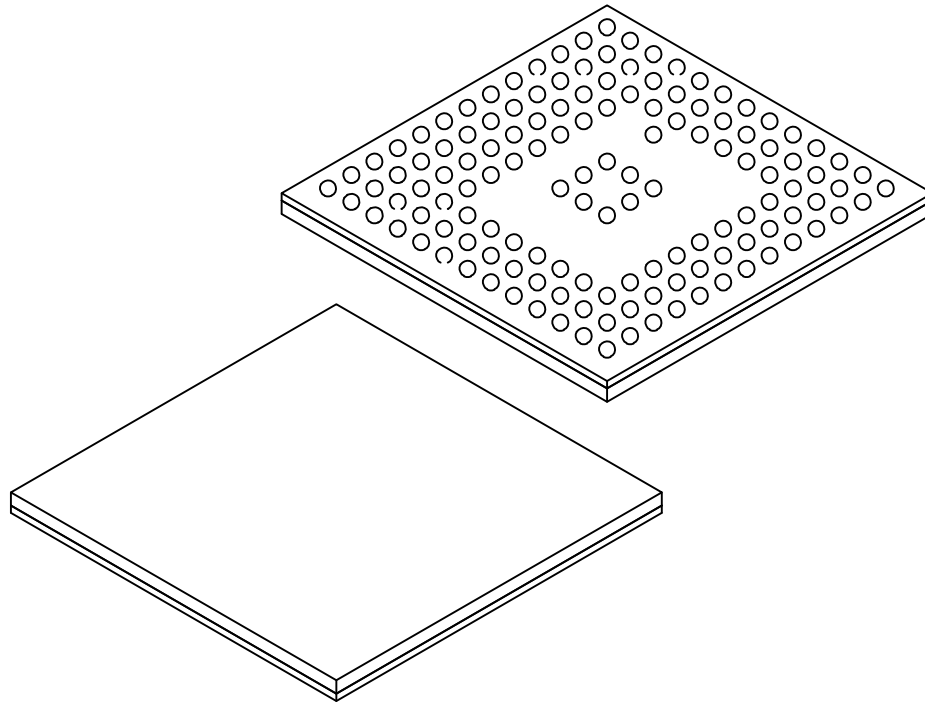
**136-ball Ultra Thin Fine-Pitch Ball Grid Array Package (C4B) - 7×7×0.6 mm Body
With 13x13 3P+7 balls+3x3 1P Array, 0.5 mm Pitch [UFBGA]; Atmel Legacy GPC CDK**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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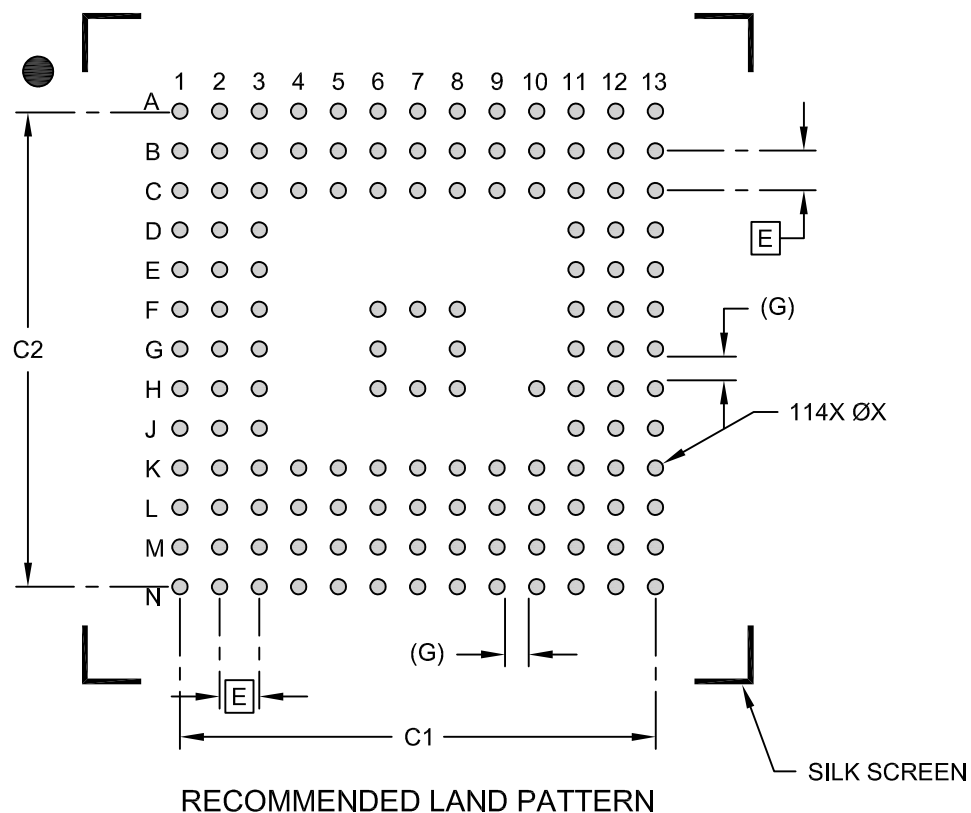
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	136		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	0.60
Ball Height	A1	0.110	-	0.210
Mold Thickness	A2	0.25 REF		
Substrate Thickness	A3	0.136 REF		
Overall Length	D	7.00 BSC		
Ball Array Length	D2	-	6.00	-
Overall Width	E	7.00 BSC		
Ball Array Width	E2	-	6.00	-
Ball Width	b	0.220	-	0.280
Ball Diameter		0.250 REF		

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

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Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Optional Pad Diameter (X114)	X			0.20
Contact Pad Spacing	C1		6.00	
Contact Pad Spacing	C2		6.00	
Column and Row Spacing	G	0.30 REF		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23165 Rev A